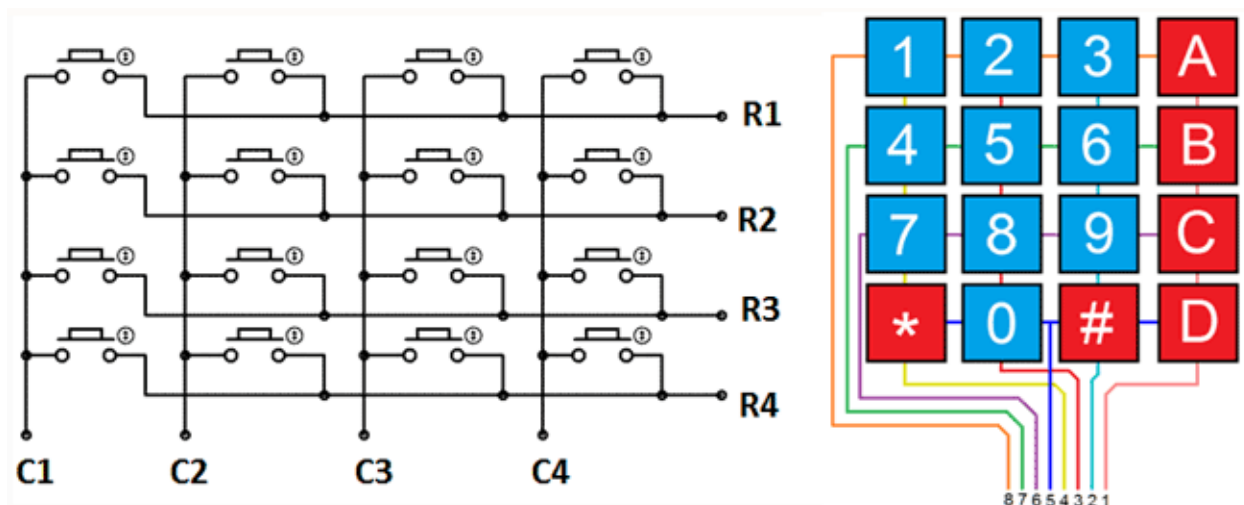
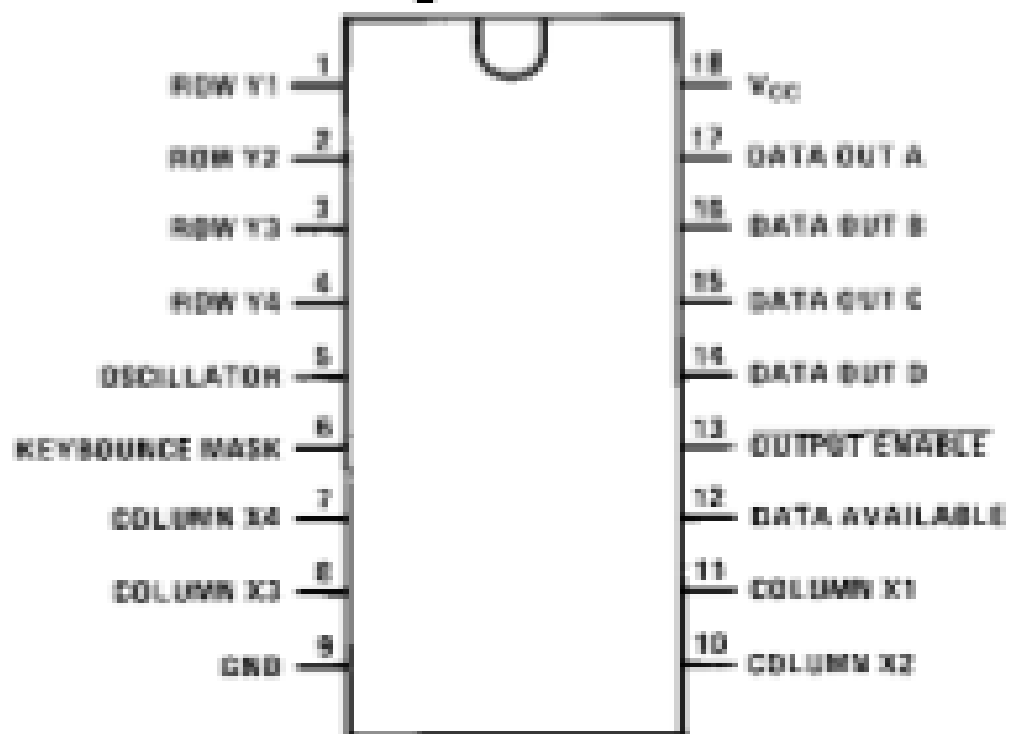


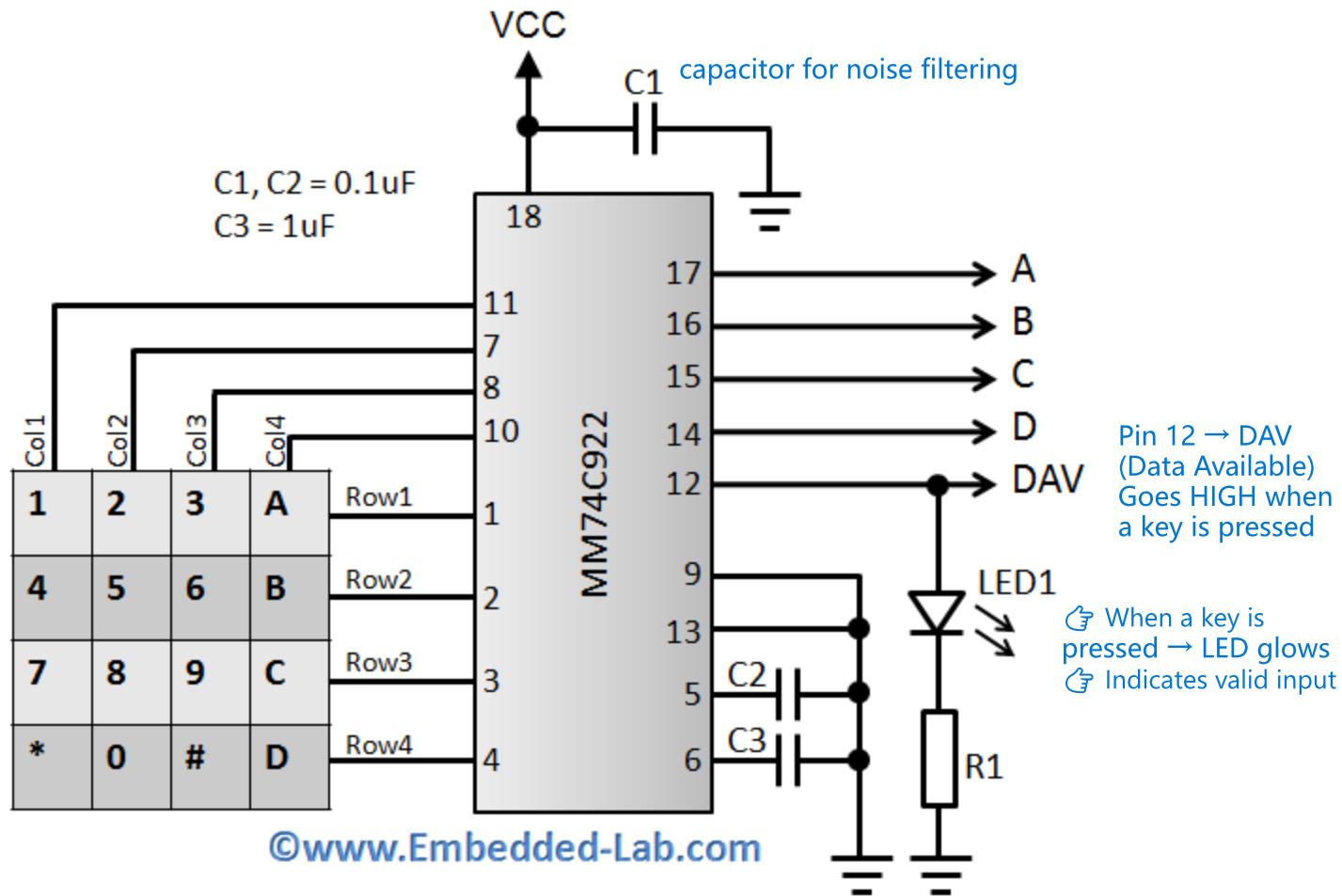


Connection Diagrams

Pin Assignment for DIP



Top View
MM74C922



encoder IC (MM74C922). The IC converts a key press into a 4-bit binary output (A, B, C, D).

Columns → connected to IC pins: 11, 7, 8, 10

Rows → connected to IC pins: 1, 2, 3, 4

When you press a key:

One row + one column get connected
The IC detects which key is pressed

Press "1" → output might be 0001

Press "5" → output might be 0101

Timing capacitors:

C2 (0.1μF) & C3 (1μF)

→ control internal timing (debouncing, scanning speed)

Scans rows and columns repeatedly

Waits a small time to confirm the key is stable (debouncing)

This "waiting time" is controlled by the capacitors.

otherwise if a signal is not clean bounces rapidly, it is not taken stable, so capacitor make sure a valid press

How does a capacitor create timing?

A capacitor:

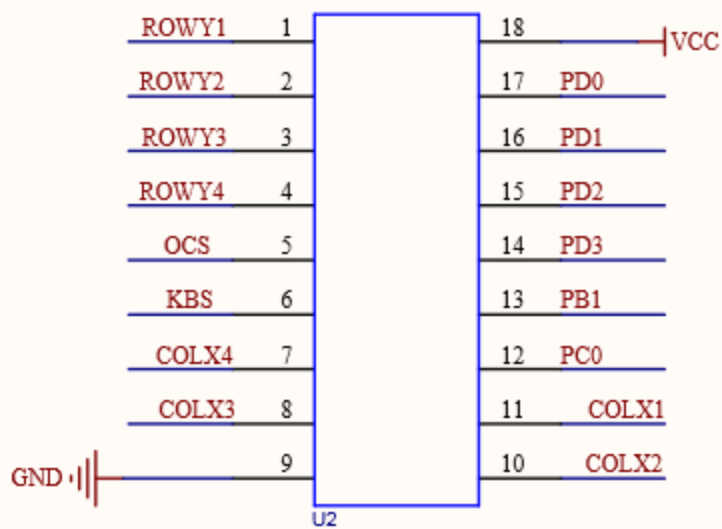
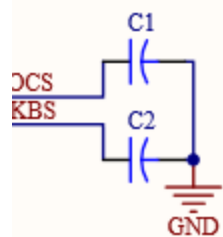
Charges and discharges slowly
This creates a time delay

☞ The bigger the capacitor:

More delay (slower operation)

☞ The smaller the capacitor:

Less delay (faster operation)



Truth Tables

(Pins 0 through 11)

Switch Position	0 Y1, X1	1 Y1, X2	2 Y1, X3	3 Y1, X4	4 Y2, X1	5 Y2, X2	6 Y2, X3	7 Y2, X4	8 Y3, X1	9 Y3, X2	10 Y3, X3	11 Y3, X4
D												
A A	0	1	0	1	0	1	0	1	0	1	0	1
T B	0	0	1	1	0	0	1	1	0	0	1	1
A C	0	0	0	0	1	1	1	1	0	0	0	0
O D	0	0	0	0	0	0	0	0	1	1	1	1
U E (Note 1)	0	0	0	0	0	0	0	0	0	0	0	0
T												

(Pins 12 through 19)

Switch Position	12 Y4, X1	13 Y4, X2	14 Y4, X3	15 Y4, X4	16 Y5 (Note 1), X1	17 Y5 (Note 1), X2	18 Y5 (Note 1), X3	19 Y5 (Note 1), X4
D								
A A	0	1	0	1	0	1	0	1
T B	0	0	1	1	0	0	1	1
A C	1	1	1	1	0	0	0	0
O D	1	1	1	1	0	0	0	0
U E (Note 1)	0	0	0	0	1	1	1	1
T								

Note 1: Omit for MM74C922